

UNIVERSITY OF MARYLAND, BALTIMORE COUNTY

**CMSC 687/487: Introduction to Network Security**

Fall 2026

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| <b>Lectures:</b>      | Monday and Wednesday, 2:30–3:45 PM                                                            |
| <b>Location:</b>      | Information Technology (ITE) 104                                                              |
| <b>Instructor:</b>    | Dr. Yiwen Hu <huyiwen@umbc.edu>                                                               |
| <b>TA:</b>            | Haitian Yan <hyan1@umbc.edu>                                                                  |
| <b>Prerequisites:</b> | Undergraduate: CMSC 421 or CMSC 481 (grade of C or better)<br>Graduate: CMSC 341 and CMSC 481 |

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## Course Description

This course covers concepts, principles, and practical techniques for securing computer networks. Major topics include applied cryptography for network protocols, Internet protocol security, network attacks and defenses (e.g., denial-of-service, spoofing, and eavesdropping), and wireless and cellular network security, as well as emerging directions such as security of AI-driven agentic network operations, space-ground integrated networks, and connected systems (IoT, V2X). Students will develop practical skills through applied exercises, gain hands-on experience with security tools and techniques, and be inspired to pursue advanced projects and research in network security.

## Office Hours

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**Instructor: Dr. Yiwen Hu**

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|--------------|--------------------------------------------------------|
| Office       | ITE 322-A                                              |
| Office hours | Monday, 3:45–4:45 PM (or by appointment)               |
| Email        | <a href="mailto:huyiwen@umbc.edu">huyiwen@umbc.edu</a> |

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**Teaching Assistant: Haitian Yan**

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| Office       | ITE 352                                            |
| Office hours | Thursday, 11:00 AM–12:00 PM<br>(or by appointment) |
| Email        | <a href="mailto:hyan1@umbc.edu">hyan1@umbc.edu</a> |

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## Textbook and Reading Materials

The textbooks below are recommended but not required. Additional reading materials may be shared during lectures to spark your interest and deepen your understanding.

- *Network Security Essentials: Applications and Standards*, William Stallings, Sixth Edition, Pearson.
- *Computer & Internet Security: A Hands-on Approach*, Wenliang Du, Third Edition, self-published, May 2022.

## Communications

I encourage all students to engage actively in classroom discussions. I value your feedback throughout the semester. You are welcome to share your thoughts during office hours or reach out to me by email.

- All course announcements, materials, and assignment submissions will be handled through **Blackboard**.
- The **Discord discussion board** is a place for general questions or clarifications. Please do not share solutions or post materials that directly reveal answers.
- You can also **email** us for questions or assistance. When you do, please start your subject line with [CMSC 687/487] and send your email to BOTH the instructor and the TA using your UMBC email account.
  - We will do our best to reply within 24 hours on weekdays. Emails that do not follow these instructions or are sent at the last minute (e.g., right before an assignment is due) may not receive a response, so please plan ahead.

## Grading

This section describes the grading scale, the grading breakdown, and related course policies.

### Grading Scale

Grades are based on the percentage of total points earned ( $X$ ). Curving is at the discretion of the instructor.

| Percentage of total points | Grade |
|----------------------------|-------|
| $90\% \leq X$              | A     |
| $80\% \leq X < 90\%$       | B     |
| $70\% \leq X < 80\%$       | C     |
| $60\% \leq X < 70\%$       | D     |
| $X < 60\%$                 | F     |

## Grading Breakdown

| Component                                | Weight      | Ind./Group  | Description                                                                                                                                                                        |
|------------------------------------------|-------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Exam                                     | 25%         | Individual  | One exam, held before Thanksgiving.                                                                                                                                                |
| Homework assignments                     | 20%         | Individual  | Three assignments consisting of conceptual questions or security tool implementations.                                                                                             |
| Course project                           | 35%         | Group (3–4) | Project proposal (5%), final presentation (10%, about 15 minutes of presentation and Q&A), and final report and deliverables (20%); graded at both the group and individual level. |
| In-class quizzes                         | 10%         | Individual  | Four short quizzes with conceptual questions or exercises.                                                                                                                         |
| Research paper presentation & discussion | 10%         | Group (3–4) | Each group selects one paper from an assigned list for a 15-minute in-class presentation and discussion.                                                                           |
| <b>Total</b>                             | <b>100%</b> |             |                                                                                                                                                                                    |

## Grading Details

- **Exam:** Details regarding the exam location, time slot, and permitted materials will be announced in class. Makeup exams will not be permitted except in extreme circumstances with valid, documented justification.
- **Course Project:** The final report must follow the [IEEE CNS conference paper format](#) and be 6–9 pages long, including references. Other detailed requirements for the proposal, the presentation, and the final report will be provided in class and must be followed. Projects must relate to network security. Only proposals approved by the instructor may proceed to subsequent stages. All students are expected to participate actively at every stage, and individual contributions must be clearly indicated in all submissions for grading.
- **In-Class Quizzes:** If attendance stays strong, I'll give advance notice of quiz dates. If it drops too low, I won't. No makeup quizzes will be allowed.
- **Deadlines and Late Submission Policy:** The due dates for assignments and the project will be announced on Blackboard, with a deadline of 11:59 PM (UMBC time) on the specified day, unless stated otherwise. Submission instructions will be provided as well. Late assignments and project submissions will be accepted with the following penalties applied to the earned score:
  - Up to 12 hours late: **–10%**
  - 12–24 hours late: **–20%**
  - 24–48 hours late: **–40%**
  - More than 48 hours late: **no credit**

## Lecture Notes (Tentative)

| Lecture                                                | Date          | Subject                                                        |
|--------------------------------------------------------|---------------|----------------------------------------------------------------|
| <b>Part I: Fundamentals</b>                            |               |                                                                |
| L01                                                    | Aug. 24, 2026 | Course Overview, Introduction to Network Security              |
| L02                                                    | Aug. 26, 2026 | Networking Fundamentals Review                                 |
| L03                                                    | Aug. 31, 2026 | Cryptographic Tools for Confidentiality and Integrity          |
| L04                                                    | Sep. 2, 2026  | Symmetric Encryption (HW1 Release)                             |
|                                                        | Sep. 7, 2026  | Labor Day, no lecture                                          |
| L05                                                    | Sep. 9, 2026  | Symmetric Encryption                                           |
| L06                                                    | Sep. 14, 2026 | Public Key Cryptography (HW1 Due)                              |
| L07                                                    | Sep. 16, 2026 | Public Key Cryptography                                        |
| L08                                                    | Sep. 21, 2026 | User Authentication                                            |
| L09                                                    | Sep. 23, 2026 | User Authentication (HW2 Release)                              |
| L10                                                    | Sep. 28, 2026 | Network Access Control                                         |
| L11                                                    | Sep. 30, 2026 | Network Access Control                                         |
| L12                                                    | Oct. 5, 2026  | Transport-Layer Security (HW2 Due)                             |
| L13                                                    | Oct. 7, 2026  | Transport-Layer Security                                       |
|                                                        | Oct. 12, 2026 | Fall Recess, no lecture                                        |
| L14                                                    | Oct. 14, 2026 | IP Security (Project Proposal Due)                             |
| L15                                                    | Oct. 19, 2026 | Exam Review                                                    |
| L16                                                    | Oct. 21, 2026 | Exam                                                           |
| <b>Part II: Advanced Topics and Research Frontiers</b> |               |                                                                |
| L17                                                    | Oct. 26, 2026 | Passive and Active Attacks                                     |
| L18                                                    | Oct. 28, 2026 | Widely-Used Network Security Tools (HW3 Release)               |
| L19*                                                   | Nov. 2, 2026  | Wireless Network Security (Wi-Fi)                              |
| L20*                                                   | Nov. 4, 2026  | Wireless Network Security (IoT, V2X)                           |
| L21*                                                   | Nov. 9, 2026  | Cellular Network Security (5G) (HW3 Due)                       |
| L22*                                                   | Nov. 11, 2026 | Cellular Network Security (NTN)                                |
| L23*                                                   | Nov. 16, 2026 | Cellular Network Security (Emerging applications and services) |
| L24*                                                   | Nov. 18, 2026 | Machine Learning for Network Security                          |
| L25*                                                   | Nov. 23, 2026 | Adversarial Machine Learning and Security of AI Systems        |
|                                                        | Nov. 25, 2026 | Thanksgiving Break, no lecture                                 |
| L26*                                                   | Nov. 30, 2026 | Security Risks of Agentic Network Operations                   |
| L27                                                    | Dec. 2, 2026  | Project Final Presentations                                    |
| L28                                                    | Dec. 7, 2026  | Project Final Presentations                                    |
| L29                                                    | Dec. 9, 2026  | Project Report Writing Instructions, Class Summary             |

\* November lectures cover advanced network security topics delivered by the instructor, with up to two student research paper presentations integrated into each session. A list of research papers will be provided for groups to select from. The papers are closely related to the topics covered in the corresponding lectures.

## Academic Integrity

This course adheres to the academic honesty policy outlined at <https://academicconduct.umbc.edu/>. The following is a brief summary of the policy:

“Academic integrity is an important value at UMBC. By enrolling in this course, each student assumes the responsibilities of an active participant in UMBC’s scholarly community in which everyone’s academic work and behavior are held to the highest standards of honesty. Cheating, fabrication, plagiarism, and helping others to commit these acts are all forms of academic dishonesty, and they are wrong. Academic misconduct could result in disciplinary action that may include, but is not limited to, suspension or dismissal.”

Please complete your own work and do not seek inappropriate assistance from others. If you are struggling with the course content, it is your responsibility to communicate with the instructor and/or TA so that we can provide assistance.

### Use of AI Tools

- **Exams and Quizzes:** The use of AI tools, including, but not limited to, language models such as ChatGPT, is strictly prohibited for exams and quizzes. There are no exceptions.
- **Homework Assignments, Course Projects, and Research Paper Presentation & Discussion:** AI tools may be used only to support your understanding (e.g., clarifying concepts), and any use of AI must be proactively disclosed in your submission. You are strictly prohibited from using AI to directly complete your work, such as, but not limited to, completing assignments, generating project topics, writing reports, preparing presentation slides, or writing code. All submissions will be checked rigorously; any prohibited or undisclosed use of AI constitutes academic dishonesty and will be handled under the academic integrity policy above. If you are unsure whether a particular use of AI violates this policy, ask the instructor or the TA before you use it.

## UMBC Policies

### Accessibility and Disability Accommodations, Guidance and Resources

Accommodations for students with disabilities are provided for all students with a qualified disability under the Americans with Disabilities Act (ADA & ADAAA) and Section 504 of the Rehabilitation Act who request and are eligible for accommodations. The Office of Student Disability Services (SDS) is the UMBC department designated to coordinate accommodations that creates equal access for students when barriers to participation exist in University courses, programs, or activities.

If you have a documented disability and need to request academic accommodations in your courses, please refer to the SDS website at [sds.umbc.edu](https://sds.umbc.edu) for registration information and office procedures.

SDS email: [disAbility@umbc.edu](mailto:disAbility@umbc.edu)

SDS phone: 410-455-2459

If you will be using SDS-approved accommodations in this class, please contact the instructor to discuss implementation of the accommodations.

### Sexual Assault, Sexual Harassment, and Gender-Based Violence and Discrimination

[UMBC Policy](#) in addition to federal and state law (to include Title IX) prohibits discrimination and harassment on the basis of sex, sexual orientation, and gender identity in University programs and activities. Any student who is impacted by sexual harassment, sexual assault, domestic violence, dating violence, stalking, sexual exploitation, gender discrimination, pregnancy discrimination, gender-based harassment, or related retaliation should contact the University's Title IX Coordinator to make a report and/or access support and resources. The Title IX Coordinator can be reached at [ecr@umbc.edu](mailto:ecr@umbc.edu) or 410-455-1717.

You can access support and resources even if you do not want to take any further action. You will not be forced to file a formal complaint or police report. Please be aware that the University may take action on its own if essential to protect the safety of the community.

If you are interested in making a report, please use the [Online Reporting/Referral Form](#). Please note that, if you report anonymously, the University's ability to respond will be limited.

**Notice that Faculty and Teaching Assistants are Mandated Reporters with Mandatory Reporting Obligations** All faculty members and teaching assistants are considered Mandated Reporters, per UMBC's [Interim Policy on Sex Discrimination, Sex-Based Harassment, and Sexual Misconduct](#). Faculty and teaching assistants are therefore required to report all known information regarding alleged conduct that may be a violation of the Policy to the Title IX Coordinator, even if a student discloses an experience that occurred before attending UMBC and/or an incident that only involves people not affiliated with UMBC. Reports are required regardless of the amount of detail provided and even in instances where support has already been offered or received.

While faculty members want to encourage you to share information related to your life experiences through discussion and written work, students should understand that faculty are required to report past and present sexual harassment, sexual assault, domestic and dating violence, stalking, and gender discrimination that is shared with them to the Title IX Coordinator so that the University can inform students of their [rights, resources, and support](#). While you are encouraged to do so, you are not obligated to respond to outreach conducted as a result of a report to the Title IX Coordinator.

If you need to speak with someone in confidence, who does not have an obligation to report to the Title IX Coordinator, UMBC has a number of [Confidential Resources](#) available to support you:

- [Retriever Integrated Health \(Main Campus\)](#): 410-455-2472; Monday–Friday 8:30 a.m.–5 p.m.; for after-hours support, call 988.
- [Center for Counseling and Well-Being](#) (Shady Grove Campus): 301-738-6273; Monday–Thursday 10:00 a.m.–7:00 p.m. and Friday 10:00 a.m.–2:00 p.m. (virtual); [Online Appointment Request Form](#)
- Pastoral Counseling via [The Gathering Space for Spiritual Well-Being](#): 410-455-6795; [i3b@umbc.edu](mailto:i3b@umbc.edu); Monday–Friday 8:00 a.m.–10:00 p.m.
- [Women, Gender, and Equity Center](#) (open to students of all genders): 410-455-2714; [womenscenter@umbc.edu](mailto:womenscenter@umbc.edu); Monday–Thursday 9:30 a.m.–5:00 p.m. and Friday 10:00 a.m.–4:00 p.m.

**Other Resources** [Shady Grove Student Resources](#), [Maryland Resources](#), [National Resources](#).

**Child Abuse and Neglect** Please note that Maryland law and [UMBC policy](#) require that faculty report all disclosures or suspicions of child abuse or neglect to the Department of Social Services and/or the police even if the person who experienced the abuse or neglect is now over 18.

## Pregnant and Parenting Students

UMBC's [Interim Policy on Sex Discrimination, Sex-Based Harassment, and Sexual Misconduct](#) expressly prohibits all forms of discrimination and harassment on the basis of sex, including pregnancy. Resources for pregnant, parenting, and breastfeeding students are available through the University's [Office of Equity and Civil Rights](#). Pregnant and parenting students are encouraged to contact the Title IX Coordinator to discuss plans and ensure ongoing access to their academic program with respect to a leave of absence—returning following leave, or any other accommodation that may be needed related to pregnancy, childbirth, adoption, breastfeeding, and/or the early months of parenting.

In addition, students who are pregnant and have an impairment related to their pregnancy that qualifies as a disability under the ADA may be entitled to accommodations through the [Office of Student Disability Services](#).

## Religious Observances & Accommodations

UMBC [Policy](#) provides that students should not be penalized because of observances of their religious beliefs, and that students shall be given an opportunity, whenever feasible, to make up within a reasonable time any academic assignment that is missed due to individual participation in religious observances. It is the responsibility of the student to inform the instructor of any intended absences or requested modifications for religious observances in advance, and as early as possible. For questions or guidance regarding religious observances and accommodations, please contact the Office of Equity and Civil Rights at [ecr@umbc.edu](mailto:ecr@umbc.edu).

## Hate, Bias, Discrimination and Harassment

UMBC values safety, cultural and ethnic diversity, social responsibility, lifelong learning, equity, and civic engagement. Consistent with these principles, [UMBC Policy](#) prohibits discrimination and harassment in its educational programs and activities or with respect to employment terms and conditions based on race, creed, color, religion, sex, gender, pregnancy, ancestry, age, gender identity or expression, national origin, veterans status, marital status, sexual orientation, physical or mental disability, or genetic information.

Students (and faculty and staff) who experience discrimination, harassment, hate, or bias based upon a protected status or who have such matters reported to them should use the [online reporting/referral form](#) to report discrimination, hate, or bias incidents. You may report incidents that happen to you anonymously. Please note that, if you report anonymously, the University's ability to respond may be limited.